VirtualPsych

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Spring 2020

A Dissertation Submitted To

Faculty of Computing,

Riphah International University, Islamabad

As a Partial Fulfillment of the Requirement for the Award of the

Degree of

Bachelors of Science in Computer Science

Faculty of Computing
Riphah International University, Islamabad

Date: [date of final presentation]

Final Approval

This is to certify that we have read the report submitted by ***Talha Islam (21688), Hamid Zaib (21684)*** for the partial fulfillment of the requirements for the degree of the Bachelors of Science in Computer Science (BSCS). It is our judgment that this report is of sufficient standard to warrant its acceptance by Riphah International University, Islamabad for the degree of Bachelors of Science in Computer Science (BSCS).

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1

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Declaration

We hereby declare that this document "**VirtualPsych**" neither as a whole nor as a part has been copied out from any source. It is further declared that we have done this project with the accompanied report entirely on the basis of our personal efforts, under the proficient guidance of our teachers, especially our supervisor **Hasnat Ali**. If any part of the system is proved to be copied out from any source or found to be reproduction of any project from anywhere else, we shall stand by the consequences.

Talha Islam

21688

Hamid Zaib

21684

Dedication

To our wonderful parents, who have always been there for us with love and support. We are forever grateful for the guidance and encouragement from our supervisor, without whom we wouldn't have been able to achieve our project goals. Thank you all for your dedication and kindness throughout this journey.

Acknowledgement

We would like to express our heartfelt gratitude to Allah Almighty for providing us with the strength and knowledge to successfully complete this project. We extend our sincere thanks to our project supervisor, Mr. Hasnat Ali, for his valuable guidance and vision. We are truly grateful for his continuous support and precious time throughout the project.

We would also like to acknowledge and thank our parents for their unwavering encouragement, support, and belief in our abilities. Their constant attention and confidence in us have been instrumental in our success.

We would like to express our appreciation to everyone who has directly or indirectly contributed to this project. Your assistance and involvement have played a significant role in our journey.

Lastly, we would like to express our gratitude to each other, Talha Islam and Hamid zaib, as a project team. Your caring support during challenging situations has been invaluable.

Thank you all for being a part of our project and for your contributions.

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Table of Contents

List of Tables

List of Figures

Abstract

VirtualPsych is a chatbot application that provides accessible and personalized mental health support to individuals. It is designed to address the increasing demand for mental health services by offering a continuous and easily accessible platform for users to discuss their mental health concerns. The app uses advanced NLP algorithms to handle open-ended conversations and offer personalized support to users. This makes it a unique and effective solution for mental health issues. With VirtualPsych, users can benefit from continuous support from the comfort of their own home, at any time. The app is easy to use and provides a simple and accessible way for individuals to get the help they need. By providing personalized support and making it easy for individuals to access help when they need it, VirtualPsych is helping to break down barriers and make mental health care more accessible to everyone. With VirtualPsych, users can feel confident that they are getting the support they need, when they need it, without having to leave their home or schedule appointments with a therapist. This makes it an ideal solution for individuals who are looking for a convenient and effective way to manage their mental health.

Chapter 1: Introduction

VirtualPsych is an innovative chatbot application designed to provide convenient and personalized assistance for mental health concerns. Mental health challenges affect a significant number of individuals globally, and unfortunately, there is often a shortage of available therapists to meet the growing demand. VirtualPsych strives to bridge this gap by offering continuous and easily accessible support to users seeking mental health assistance.

The main challenge that our project addresses is the current limitations of existing chatbots in the mental health field. While chatbots have gained popularity, they often struggle with engaging in open-ended conversations and fail to deliver personalized support tailored to each user's specific mental health requirements. Therefore, there is a pressing need for an advanced chatbot that can adeptly handle open-ended or close to open-ended discussions and offer individualized assistance to users.

Through VirtualPsych, users can engage in meaningful conversations about their mental well-being, expressing their thoughts, feelings, and concerns openly. The chatbot is designed to be highly interactive, providing a supportive space where users can discuss their experiences without fear of judgment.

Furthermore, VirtualPsych goes beyond mere conversation by incorporating effective meditation practices into its services. The chatbot guides users through various meditation techniques that promote relaxation, mindfulness, and self-reflection. These meditation processes are thoughtfully integrated to help individuals find their inner peace, reduce stress, enhance focus, and cultivate a positive mindset.

By utilizing VirtualPsych, individuals can conveniently access mental health support whenever they need it, even in the absence of readily available therapists. The chatbot's ability to engage in open-ended conversations and tailor its responses to the unique needs of each user ensures a personalized and empowering experience.

In conclusion, VirtualPsych aims to revolutionize the way mental health support is delivered by offering a user-friendly chatbot that excels in open-ended discussions and provides personalized assistance. By combining the power of conversation and meditation, VirtualPsych strives to improve the mental well-being of individuals and create a more accessible and inclusive mental health support system

1.1 Goals and Objectives

The goal of this project is to design and develop a chatbot application called VirtualPsych that functions as a virtual psychologist, providing personalized support to users experiencing mental health issues. The objectives of the project are:

- To create an advanced chatbot that can handle open-ended conversations and provide personalized support tailored to the user's specific mental health needs.
- To use natural language processing and machine learning techniques to understand and respond to user messages.
- To offer continuous and accessible mental health support to users, filling the gap created by the limited availability of therapists.

1.2 Scope of the Project

The VirtualPsych chatbot app project aims to design and develop a user-friendly application that provides personalized mental health support. One of the primary objectives is to design the chatbot app in a way that enables it to engage in open-ended conversations. This means that users can freely express their thoughts, feelings, and concerns without feeling limited by predefined options. The chatbot will be equipped to understand and respond appropriately to the messages received from users, ensuring a meaningful and supportive interaction.

Additionally, the app will offer a diverse range of guided meditation sessions. Guided meditation involves following audio or visual cues to help individuals relax and focus their minds. The chatbot will provide access to these sessions, making it easier for users to incorporate meditation into their daily routine for relaxation and emotional well-being.

The goal is to create a user-friendly interface that is easy to navigate, ensuring that users can effortlessly engage in meaningful conversations, access valuable mental health resources, and participate in guided meditation sessions. The app will prioritize simplicity and accessibility, making it suitable for individuals of all ages and backgrounds seeking mental health support.

In summary, VirtualPsych is an innovative chatbot application that aims to provide convenient and personalized mental health assistance. With the increasing prevalence of mental health challenges and a shortage of available therapists, VirtualPsych fills the gap by offering continuous and easily accessible support to users. The main challenge addressed by the project is the limitations of existing chatbots in handling open-ended conversations and providing personalized support. VirtualPsych provides a supportive space for users to openly express their thoughts and concerns about their mental well-being. The chatbot also incorporates effective meditation practices to promote relaxation, mindfulness, and self-reflection. By combining conversation and meditation, VirtualPsych offers a user-friendly and personalized experience, revolutionizing the way mental health support is delivered. The project goals include developing an advanced chatbot, utilizing natural language processing and machine learning techniques, and providing continuous and accessible mental health support. The scope of the project encompasses designing a user-friendly interface for open-ended conversations, offering diverse guided meditation sessions, and prioritizing simplicity and accessibility for users seeking mental health support.

Chapter 2: Literature Review

2.1 Background and Problem Elaboration

2.2.1 Background

The field of mental health has long been associated with face-to-face interactions between individuals and healthcare professionals.

- Traditional mental health support faces challenges of limited accessibility, high costs, long waiting times, and stigma.
- Virtual psychologist chatbots offer a solution by leveraging AI technologies.
- Chatbots engage in conversational interactions and provide mental health services.

To address these challenges and provide accessible and convenient mental health support, virtual psychologist chatbots have emerged as a promising solution. These chatbots leverage artificial intelligence (AI) technologies, such as natural language processing and machine learning, to engage in conversational interactions with users, offering a range of mental health services.

2.2.2 Problem Elaboration

Despite the potential benefits of virtual psychologist chatbots in providing accessible and convenient mental health support, there are several challenges that need to be addressed:

1. **Limited Accessibility:** Traditional mental health support often faces limitations in terms of accessibility. Many individuals may find it challenging to access in-person therapy due to factors such as geographical barriers, transportation limitations, or a lack of available mental health professionals in their area. Virtual psychologist chatbots aim to bridge this accessibility gap by providing support anytime, anywhere, through a digital platform.
2. **High Costs:** In-person therapy sessions can be costly, making mental health support unaffordable for many individuals. Virtual psychologist chatbots offer a cost-effective alternative by providing free or affordable mental health services. This enables individuals who may not have the financial means to access traditional therapy to still receive support and guidance.
3. **Long Waiting Times:** Another challenge in traditional mental health support is the long waiting times to see a healthcare professional. This delay can exacerbate mental

health issues and discourage individuals from seeking help. Virtual psychologist chatbots offer immediate support, allowing users to receive assistance without the need for appointments or waiting periods.

4. **Stigma and Privacy Concerns:** The stigma associated with seeking mental health support can be a significant barrier for many individuals. Virtual psychologist chatbots provide a confidential and judgment-free environment where users can discuss their concerns without fear of stigma or social repercussions. Privacy concerns are also addressed by ensuring secure data handling and maintaining user confidentiality.
5. **Accuracy and Reliability of Responses:** To gain users' trust and provide effective support, virtual psychologist chatbot must deliver reliable responses. The challenge lies in training the AI algorithms to understand and interpret users' input correctly and provide appropriate and evidence-based recommendations or strategies.

2.2 Detailed Literature Review

2.2.1 Definitions

Before delving into the related research work, it is important to define some key terms relevant to virtual psychologist chatbots:

Virtual Psychologist Chatbot: A virtual psychologist chatbot refers to an AI-powered conversational agent designed to simulate human-like conversations with users, providing mental health support, guidance, and resources.

Natural Language Processing (NLP): Natural language processing is a subfield of AI that focuses on the interaction between computers and human language. It enables chatbots to understand, interpret, and generate human language, allowing for meaningful and contextually appropriate conversations.

Machine Learning (ML): Machine learning is a branch of AI that involves the development of algorithms and models that enable computers to learn from data and make predictions or decisions without explicit programming. ML techniques are often used in chatbots to improve their performance and adaptability.

Guided Meditation Sessions: Guided meditation Sessions is a practice that involves the use of verbal instructions or audio recordings to lead individuals through a meditation session. It typically includes relaxation techniques, mindfulness exercises, and visualization to promote relaxation, stress reduction, and mental clarity. Guided meditation sessions are often designed to address specific concerns such as anxiety, sleep issues, or improving focus.

User-Friendly Application: A user-friendly application is designed with a focus on simplicity, intuitiveness, and ease of use for the end-users. It prioritizes clear navigation, intuitive interfaces, and straightforward functionalities to enhance the user experience. In the context of our project, the VirtualPsych chatbot app aims to provide a user-friendly interface that allows users to access mental health support and engage in guided meditation seamlessly.

2.2.2 Related Research Work 1

Woebot is an AI-powered chatbot designed to provide mental health support, particularly in the area of cognitive-behavioral therapy (CBT). The methodology employed in the development of Woebot involves the use of NLP techniques and cognitive restructuring strategies.

In a notable study by Fitzpatrick [1] the effectiveness of Woebot as a virtual psychologist chatbot was examined in individuals with anxiety disorders. The study utilized a randomized controlled trial methodology to assess the outcomes of the intervention. Participants were assigned to either the Woebot intervention group or a control group.

The NLP component of Woebot involved processing and parsing user input to extract relevant information and identify cognitive distortions. This enabled Woebot to understand and analyze the user's thoughts and provide appropriate responses. The chatbot utilized a rule-based algorithm to generate responses and engage users in CBT techniques such as cognitive restructuring and thought challenging.

The study findings demonstrated promising outcomes, showing that Woebot was effective in reducing anxiety symptoms and improving overall well-being in the intervention group. This research highlights the potential of virtual psychologist chatbots, like Woebot, in delivering accessible and effective mental health support using CBT principles and NLP techniques.

2.2.3 Related Research Work 2

In a study conducted by Prabod Rathnayaka et al. [2], a BA-based AI chatbot was developed and implemented as a cross-platform smartphone application. The application was built using the Python programming language for the backend and the JavaScript programming language for the frontend. The research paper provides a detailed description of the technical architecture used in the implementation.

The smartphone application was developed using the react native library, allowing it to be compiled for both Android and iOS operating systems. The backend application consisted of three main components: REST Programming Interface (API), back office, and AI engine. The REST API facilitated communication between the mobile application and the backend and was implemented using the Flask framework.

The feature extractor utilized a lexical syntactic featurizer and count vector featurizer to extract multiple sparse and dense features from user messages. The lexical syntactic featurizer generated lexical and syntactic features to support entity extraction, while the count vectorizer created bag-of-words representations of user messages, intents, and responses. Additionally, word embeddings were extracted as features using the BERT Language model. This comprehensive feature extraction process enhanced the chatbot's understanding and response generation capabilities. The research demonstrated the successful implementation of a scalable chatbot for personalized behavioral activation and remote health monitoring.

2.2.4 Related Research Work 3

The research paper "Therapy Chatbot: A Relief From Mental Stress And Problems" [3] explores the development of a therapy chatbot to address mental health issues and provide support to individuals. It emphasizes the importance of mental health in today's society, highlighting the need for individuals to have a safe and non-judgmental space to express their emotions and seek assistance. The proposed therapy chatbot aims to fulfill this need by utilizing natural language processing (NLP) techniques and a Sequence to Sequence architecture implemented in the DialogFlow framework. Through supervised learning, the chatbot can generate appropriate responses based on user input, covering a range of situations such as personal loss, relationship problems, career/job issues, and academic/education-related concerns. The chatbot is integrated into a Flutter app, ensuring accessibility on Android devices.

The research work utilizes a basic Sequence to Sequence architecture in building the conversational therapy chatbot. The development is carried out using the DialogFlow framework, a platform that provides natural language processing capabilities. Within the architecture, different entities are created to cater to specific situations, enabling the chatbot to provide relevant responses. The training data for the chatbot is obtained through supervised learning, allowing the bot to detect keywords and generate appropriate replies. To ensure usability and accessibility, the chatbot is implemented as an app using Flutter, a mobile UI framework. The app's user interface facilitates user input, which is then sent to the chatbot application. The chatbot performs preprocessing steps such as tokenization, stop word removal, and highlighting, and the question and answer pairs are stored in a database for retrieval when matching queries are encountered. Overall, this research methodology leverages NLP techniques, utilizes the DialogFlow platform, and employs supervised learning to develop a therapy chatbot that can effectively address mental health concerns and provide support to individuals in a user-friendly manner.

2.3 Literature Review Summary Table

Table 1: Literature Review Summary Table

No.	Application Name	Inventor	Methodology	Key Findings
1	Eliza	Joseph Weizenbaum	Rule-Based Algorithm	Eliza, an early chatbot, demonstrated the ability to engage users in conversations and simulate human-like interactions.
2	Woebot	Dr. Alison Darcy	Rule-Based Algorithm	Woebot, an AI-powered chatbot utilizing NLP and CBT techniques, showed promising outcomes in reducing anxiety symptoms and improving well-being.
3	BA-based AI Chatbot Bunji	La Trobe University	Bert and Rasa	A BA-based AI chatbot implemented as a cross-platform smartphone application demonstrated successful scalability for personalized behavioral activation.
4	Wysa	Touchkin	AI-powered chatbot with rule based decision making	Emotional support, CBT techniques, mood tracking, guided meditation, coping strategies
5	Therapy chatbot	Zeeshan Ahmad	Sequence to Sequence Architecture	The therapy chatbot utilizes a sequence to sequence architecture to provide a non-judgmental platform for individuals to express their emotions, reducing stress and improving productivity.

Research Gap

The literature review identified several gaps in the existing research. These include:

- Lower accuracy: Many existing chatbot systems struggle with providing accurate and relevant responses, especially when faced with complex or open-ended questions related to mental health. These limitations hinder the effectiveness of chatbots as reliable mental health support tools.
- Inability to answer open-ended questions: Most chatbots are designed to provide predefined responses based on a set of predetermined questions and answers. However, they often lack the ability to handle open-ended questions and engage in meaningful conversations. This limitation restricts the chatbot's ability to address the diverse and individualized needs of users seeking mental health assistance.
- Lack of conversation memory: Another notable gap is the chatbot's inability to remember past conversations with users. Without the capability to recall previous interactions, chatbots may struggle to maintain continuity in conversations and provide personalized support based on the user's history. This limitation hampers the development of a more personalized and tailored mental health assistance experience.
- Rule-based nature of existing chatbots: Most mental health chatbots rely on rule-based approaches, where predefined rules and decision trees govern their responses. This restricts their ability to adapt to individual users' needs and preferences and limits the chatbot's capacity to provide personalized and dynamic support.

2.4 Problem Statement

The aim is to address the existing limitations in terms of accuracy and human-like responses in chatbot interactions, while providing effective mental health assistance. The app will strive to improve user satisfaction and engagement by delivering empathetic and contextually relevant responses, thereby enhancing the overall user experience and mental health outcomes. Additionally, the app will provide 24/7 support to users, ensuring continuous access to mental health assistance. It will also include meditation features to support users in managing stress, promoting relaxation, and enhancing their well-being.

Summary

In this Chapter, the literature review explores the background and problem elaboration related to psychologist chatbots as a solution for accessible and convenient mental health support. It discusses the challenges faced by traditional mental health support, such as limited accessibility, high costs, long waiting times, stigma, and privacy concerns. The review highlights the emergence of virtual psychologist chatbots that leverage AI technologies, including natural language processing and machine learning, to provide mental health services through conversational interactions.

The chapter defines key terms, including virtual psychologist chatbot, natural language processing (NLP), machine learning (ML), guided meditation sessions, and user-friendly application. It emphasizes the importance of these concepts in the development and functionality of the VirtualPsych chatbot app.

Additionally, the literature review presents two related research works. The first work discusses Woebot, an AI-powered chatbot that focuses on cognitive-behavioral therapy (CBT) and demonstrates promising outcomes in reducing anxiety symptoms. The second work describes a BA-based AI chatbot implemented as a smartphone application, showcasing successful scalability for personalized behavioral activation.

The review identifies research gaps, including lower accuracy, the inability to answer open-ended questions, lack of conversation memory, and the rule-based nature of existing chatbots. These gaps highlight the need for an improved chatbot solution that addresses these limitations and provides accurate, personalized, and dynamic mental health assistance.

In summary, of this Chapter provides a comprehensive overview of the existing literature on virtual psychologist chatbots, emphasizing the challenges, potential benefits, and research gaps. It sets the foundation for the development of the VirtualPsych chatbot app, which aims to overcome these limitations by delivering accurate, human-like responses, ensuring user satisfaction, and offering 24/7 support. The incorporation of guided meditation features further enhances the app's ability to support stress management, relaxation, and overall well-being.

Chapter 3: Requirements and Design

3.1 Requirements

The requirements section outlines the functional and non-functional requirements for the virtual psychology application. Functional requirements describe the specific features and functionality that the application must have, while non-functional requirements describe the qualities or characteristics that the application must possess, such as reliability, security, and performance.

3.1.1 Functional Requirements

FR.1: Users should be able to create an account and login to the app.

FR.2: The app should have a chatbot that can interact with the users and provide them with psychological support and advice.

FR.3: App should provide personalized recommendations to users.

FR.4: Users shall be able to provide feedback on the app.

FR.5: Offer appropriate guidance and advice to user about meditation.

FR.6: Implement a feature to track and display user progress over time.

3.1.2 Non-Functional Requirements

N-FR.1: The app shall be designed to handle a large number of users.

N-FR.2: The app shall be compatible with multiple devices.

N-FR.3: The app shall have a user-friendly interface, with easy navigation and intuitive design.

3.1.3 Hardware and Software Requirements

To use the virtual psych app, users need a device that meets the minimum hardware and software requirements. The app is designed to be lightweight, which means it can run smoothly on most modern smartphones, tablets and browser.

Hardware Requirements:

HR.1: Minimum 2GB RAM (recommended 4GB or higher)

HR.2: Minimum 1GHz processor (recommended 1.8GHz or higher)

HR.3: Minimum 20MB storage space (recommended 50 MB or higher)

Software Requirements:

SR.1: (Operating system) Android 5.0 or higher / iOS 10 or higher

SR.2: (Internet connection) Minimum 3G (recommended 4G or higher)

SR.3: (Browser support) Latest version of Chrome, Firefox, Safari, or Edge

Table 2: Software and Hardware requirments

Requirement	Minimum	Recommended
Hardware Requirements		
RAM	Minimum 2GB	Recommended 4GB or higher
Processor	Minimum 1GHz	Recommended 1.8GHz or higher
Storage Space	Minimum 20MB	Recommended 50MB or higher
Software Requirements		
Operating System	Android 5.0 or higher	iOS 10 or higher
Internet Connection	Minimum 3G	Recommended 4G or higher
Browser Support	Latest version of Chrome, Firefox, or Edge	

3.2 Proposed Methodology

The proposed methodology for the development of the VirtualPsych chatbot application involves several key stages to ensure the design, development, and evaluation of an effective and user-friendly virtual psychologist. The methodology includes the collection of a dataset, design and development of the chatbot architecture, and testing and evaluation of the app's usability and effectiveness.

3.2.1 Dataset Collection:

The first stage of the methodology focuses on the collection of a comprehensive dataset comprising patient and doctor interactions. This dataset will serve as the foundation for training the chatbot and enabling it to understand and respond to user messages accurately.

The dataset collection stage involved the creation of our own dataset comprising patient and psychologist interactions. We recognized the importance of having a dataset that accurately represents the real-world scenarios and challenges faced by individuals seeking mental health support. we sought validation and verification of our dataset from the reputable government hospital Benazir Bhutto Hospital. Their expert mental health professionals reviewed the dataset and confirmed its alignment with real-world mental health scenarios. Their validation provided assurance regarding the dataset's quality and suitability for training the VirtualPsych chatbot.

By creating our own dataset and verifying it through collaboration with a reputable government hospital, we aimed to ensure that the VirtualPsych chatbot is trained on a reliable and representative dataset that accurately reflects the challenges and interactions experienced in actual mental health settings.

3.2.2 Design and Development:

Once the dataset is collected and approved, the design and development stage begins. This stage involves the creation of a chatbot architecture capable of effectively interpreting user messages and providing personalized responses. To develop the mobile application, we will utilize the Flutter framework, a popular cross-platform development framework known for its fast and efficient app development capabilities. Flutter will enable us to create a user-friendly and visually appealing interface for VirtualPsych.

For the chatbot functionality, we will employ the RASA chatbot framework. RASA is an open-source framework that provides powerful tools and libraries for building conversational AI applications. It offers natural language understanding (NLU) and dialogue management capabilities, allowing us to train the chatbot to understand user intents and generate appropriate responses. RASA will enable us to incorporate advanced NLP techniques and machine learning algorithms to enhance the chatbot's conversational abilities and provide a more personalized and engaging experience for users.

3.2.3 Testing and Evaluation:

In the final stage of the methodology, rigorous testing and evaluation will be conducted to assess the usability and effectiveness of the VirtualPsych chatbot application. Usability testing will involve gathering feedback from a group of users to evaluate the app's intuitiveness, ease of use, and overall user satisfaction. Additionally, the app's effectiveness in reducing symptoms of anxiety and depression will be evaluated through user surveys and feedback from mental health professionals. This feedback will be utilized to iteratively improve the app's performance and enhance the user experience.

By following this proposed methodology, the VirtualPsych chatbot application aims to deliver personalized mental health support to users in a user-friendly and accessible manner. The utilization of natural language processing and machine learning techniques will enable the chatbot to understand and respond accurately to user messages. The continuous availability of the app will ensure users have access to support whenever they need it, contributing to their overall well-being and mental health.

3.3 System Architecture

The system architecture of our app follows a client-server model, with the client being the user-facing frontend application and the server handling the backend operations.

Client-Side:

- **Frontend Application:** The user interacts with the app through the frontend interface, which includes the chatbot interface, user input forms, and display of information.
- **User Interface (UI):** The UI is responsible for presenting the app's features, capturing user input, and displaying responses from the chatbot.
- **User Input Handling:** User input is captured and processed by the frontend, which sends requests to the server for processing and obtaining responses.

Server-Side:

- **Backend:** The backend handles the processing of user requests, performs the necessary computations, and generates appropriate responses.
- **Natural Language Processing (NLP):** The NLP component processes user input, applies language understanding techniques, and extracts relevant information.
- **Database:** The database stores user information, chat history, and for frontend storing temporarily the chat history so it may visible to user what user says before to chatbot
- **Communication:** Client-Server Communication: The client (frontend) communicates with the server (backend) through API calls, sending user input and receiving responses.
- **Database Interaction:** The backend interacts with the database to store and retrieve user-related data and chat history.

This system architecture ensures smooth communication between the user interface, the backend application, and the necessary components such as NLP and the database.

3.4 Use Cases

3.4.1 User Registration

Table 3: User Registration

Name	User Registration		
Actors	User		
Summary	The user shall provide their name, email and password on the registration form and after successful registration, the user can access the chatbot application.		
Pre-Conditions	The user must not be registered in the system.		
Post-Conditions	The user's registration information is successfully stored in the database, and the user can access the chatbot application.		
Special Requirements	None		
Basic Flow			
Actor Action		**System Response**	
1	The user opens the registration page.	2	The registration page is displayed, prompting the user to enter their email and password.
3	The user enters valid email and password.	4	The system validates the entered information, creates a new user record in the database, and displays a success message.
Alternative Flow			
3	The user enters an email or password that is already registered in the system.	4-A	The system responds with an error message: "The email or password is already registered. Please try again with a different email or password."

3.4.2 User Login

Table 4: User Login

Name	User Login		
Actors	User		
Summary	The user shall provide their email and password on the login form and after successful verification, redirect the user to the home page.		
Pre-Conditions	The user must be in the database records either added by any of the authorized users or added manually by a developer. The user must not already be logged in.		
Post-Conditions	The user's session is successfully established and shall be redirected to the home page.		
Special Requirements	None		
Basic Flow			
Actor Action		**System Response**	
1	The user opens the login page.	2	The login page is displayed asking for email and password.
3	The user enters valid email and password.	4	The system verifies the email and password, establishes a session for the user and redirects the user to the home page.
Alternative Flow			
3	The user enters invalid email or password.	4-A	The system responds with an error message: *Incorrect email or password entered.*

3.4.3 Start Conversation

Table 5: Start Conversation

Name	Start Conversation
Actors	User
Summary	The user initiates a conversation with the virtual psychologist by sending their first message.
Pre-Conditions	The user must be logged in and have access to the chatbot.
Post-Conditions	The conversation is successfully started, and the user's message is sent to

	the virtual psychologist.		
Special Requirements	None		
Basic Flow			
Actor Action		System Response	
1.	The user enters their message in the chat interface	2.	The system receives the user's message and sends it to the virtual psychologist for processing.

2.4.4 Track Task Progress

Table 6: Track Task Progress

Name	Track Task Progress		
Actors	User		
Summary	The user can track the progress of tasks they need to complete within the virtual psychologist app.		
Pre-Conditions	The user must be logged in and have an active task list.		
Post-Conditions	The user receives an overview of their task progress.		
Special Requirements	None		
Basic Flow			
Actor Action		**System Response**	
1.	The user navigates to the task progress section of the app	2.	The system retrieves the user's task list and displays an overview of task progress.
Alternative Flow			
1.	The user marks a task as complete	2-A.	The system updates the task progress and reflects the completed task in the overview.
3.	The user adds a new task to the list.	4-A	The system updates the task progress and includes the new task in the overview.

3.4.4 Access Resources

Table 7: Access Resources

Name	Access Resources		
Actors	User		
Summary	The user can access educational resources and materials.		
Pre-Conditions	The user must be logged in.		
Post-Conditions	The user can view and download the available resources.		
Special Requirements	None		
Basic Flow			
Actor Action		**System Response**	
1.	The user navigates to the resources section of the app.	2.	The system displays a list of available resources.
3.	3. The user selects a specific resource to view.	4.	The system provides the user with the selected resource.

3.5 Sequence diagram *(Optional)*

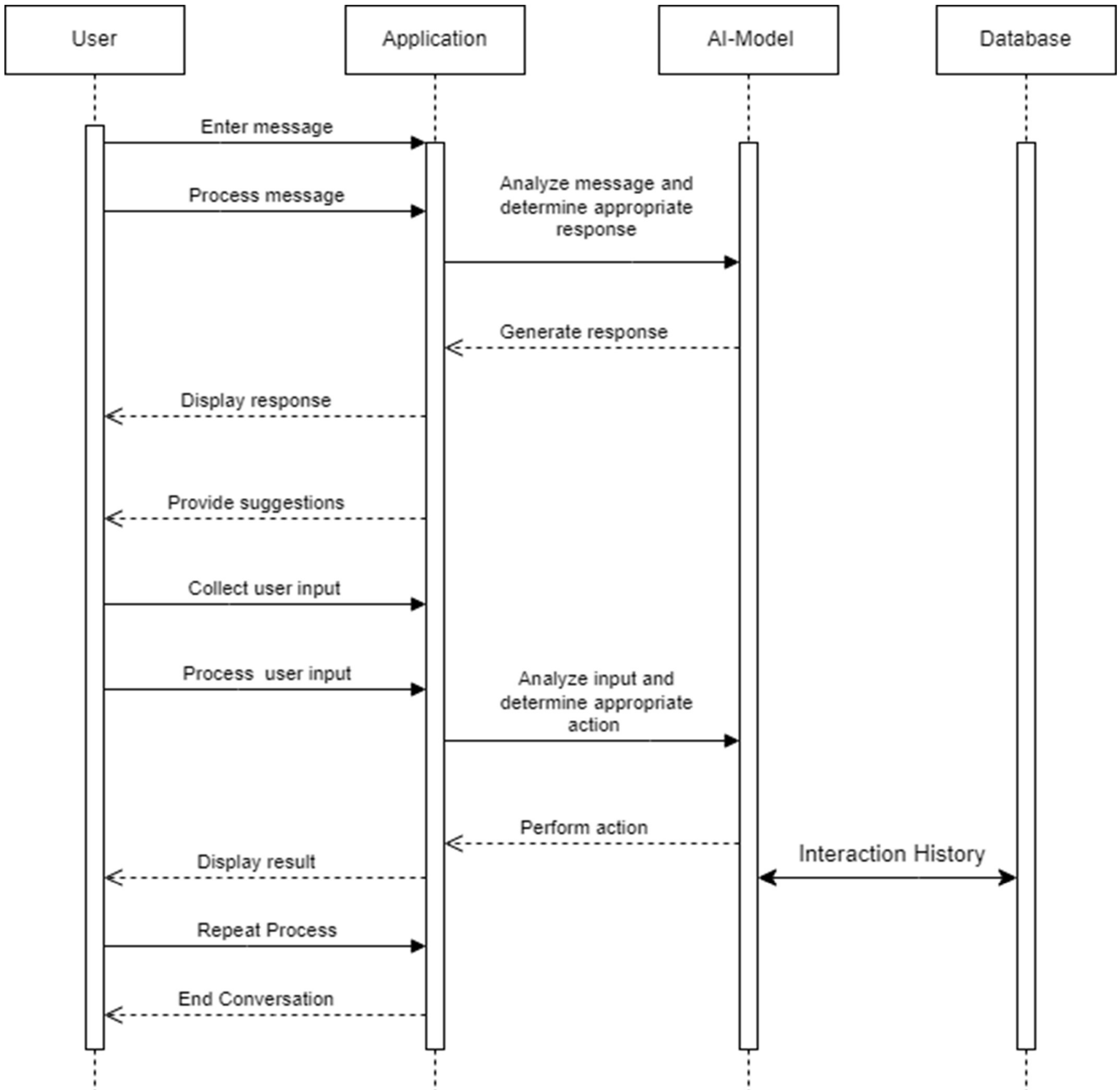


Figure 1: Sequence Diagram

3.6 Anatomy Diagram

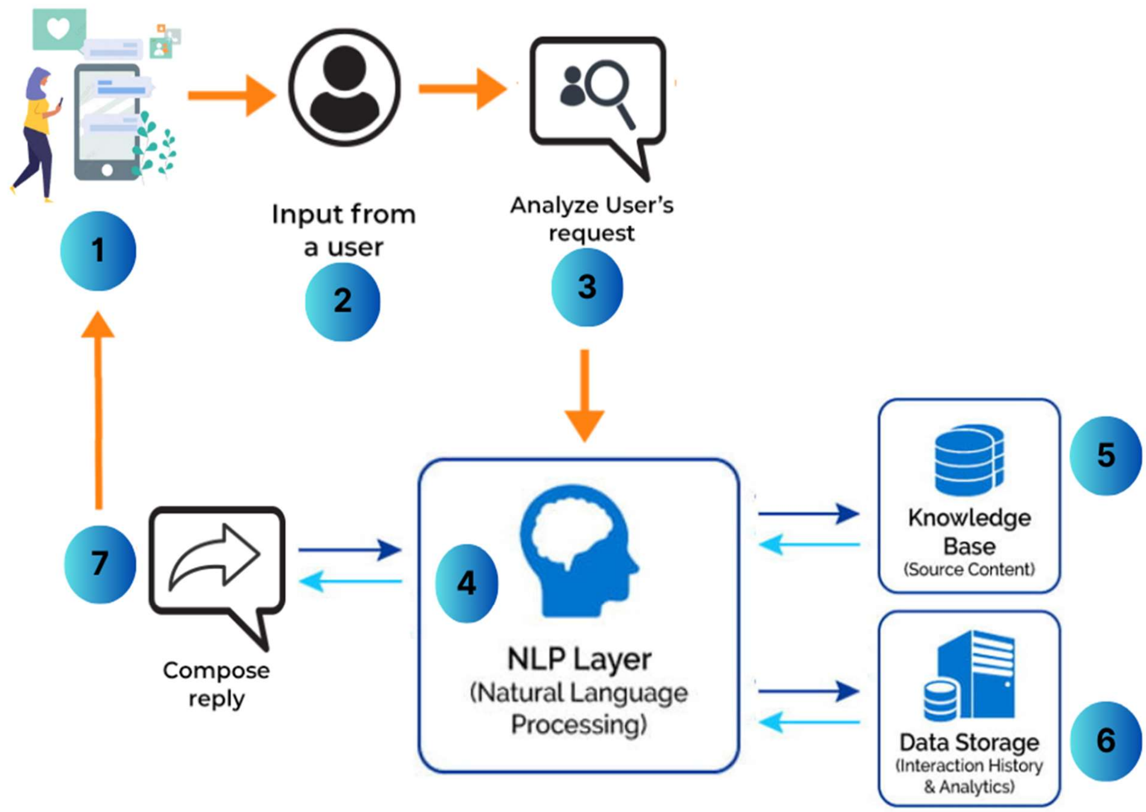


Figure 2: Anatomy Diagram

3.7 Architectural Diagram

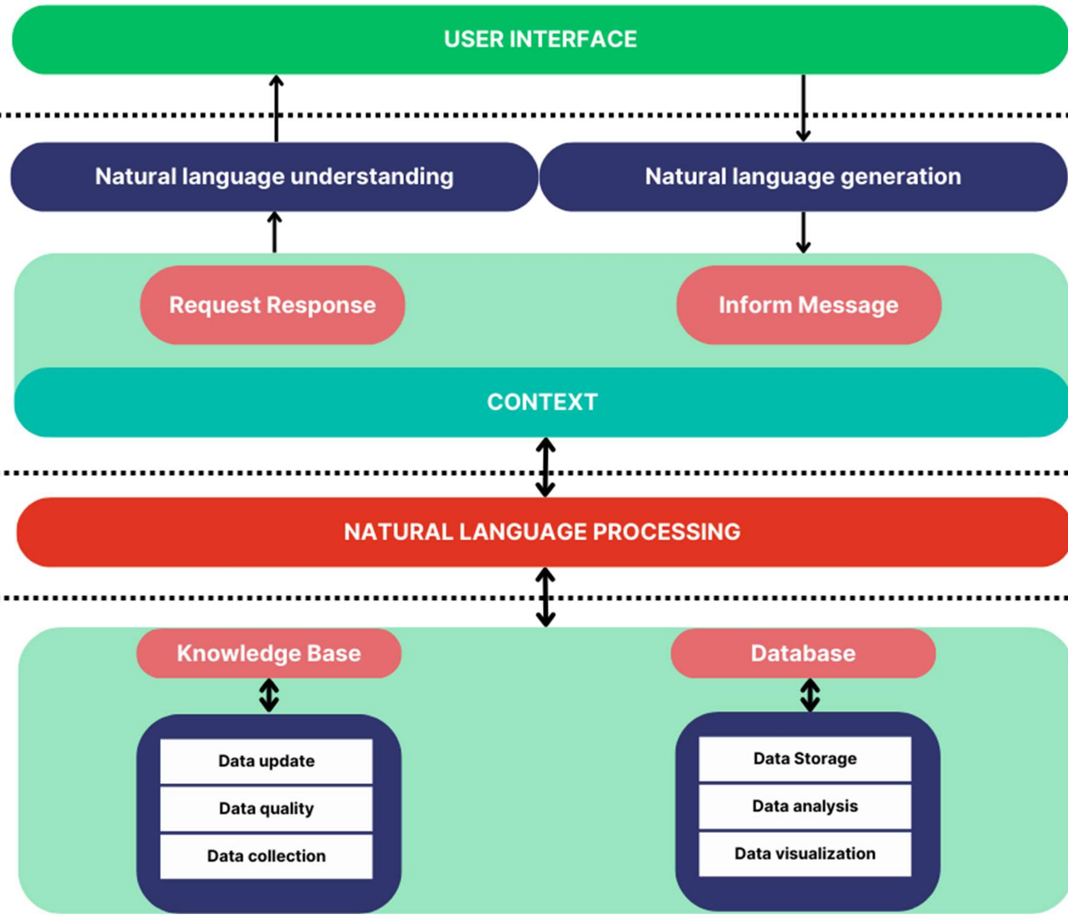
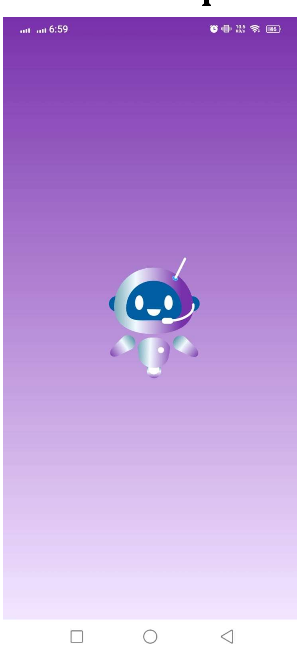
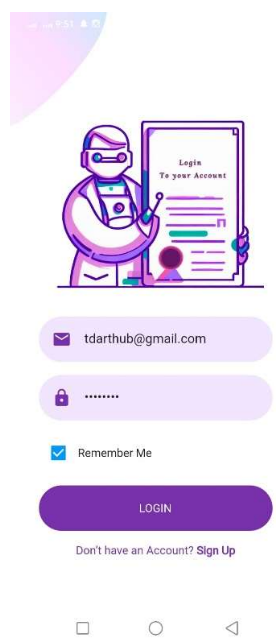
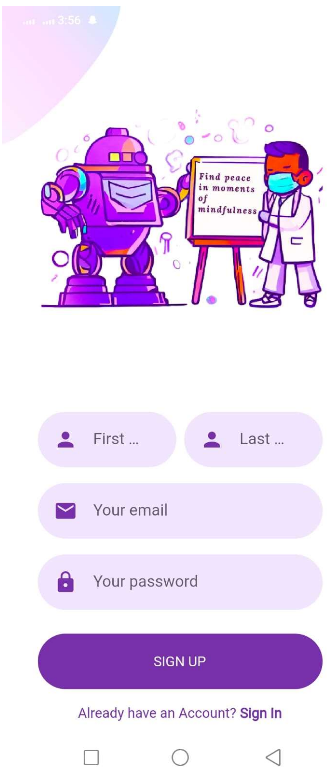


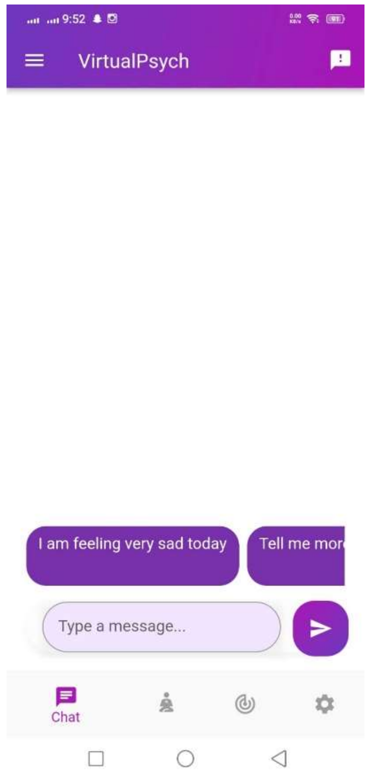
Figure 3: Architectural Diagram

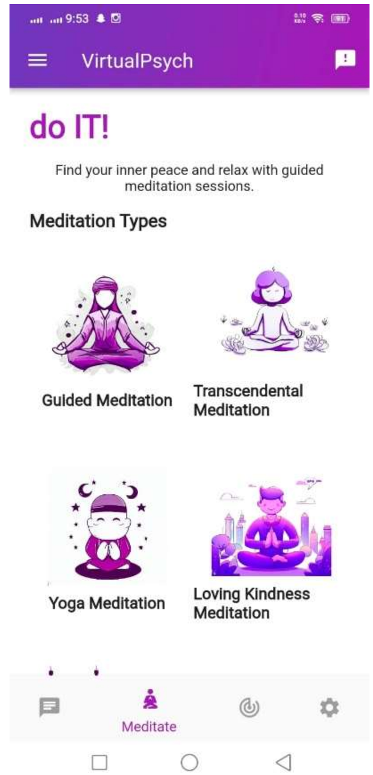
3.8 GUI Graphical User Interfaces

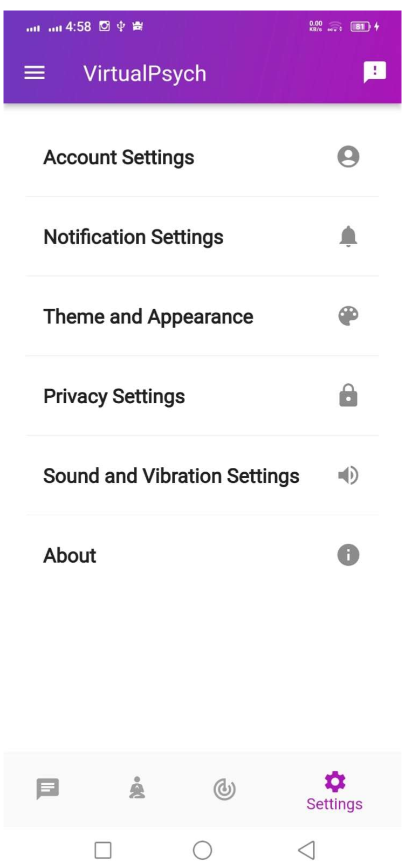


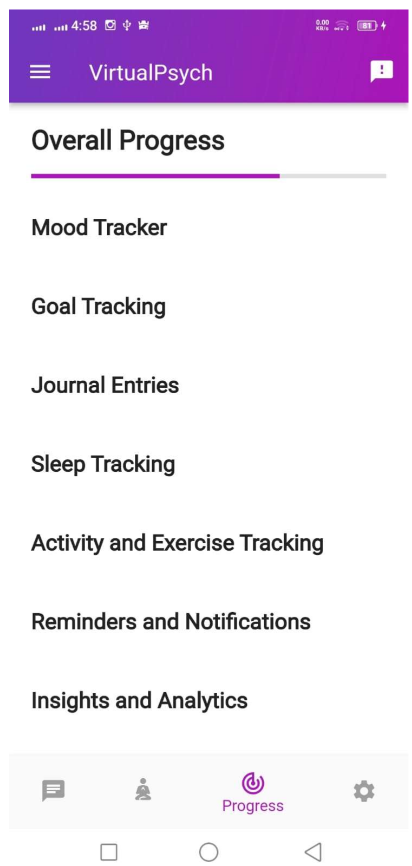


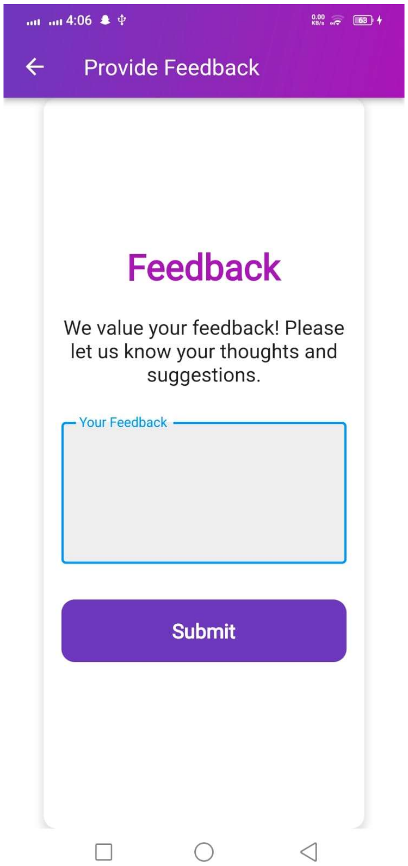

Find peace
in moments
of
mindfulness
First ...
Last ...
Your email
Your password
SIGN UP
Already have an Account? Sign In

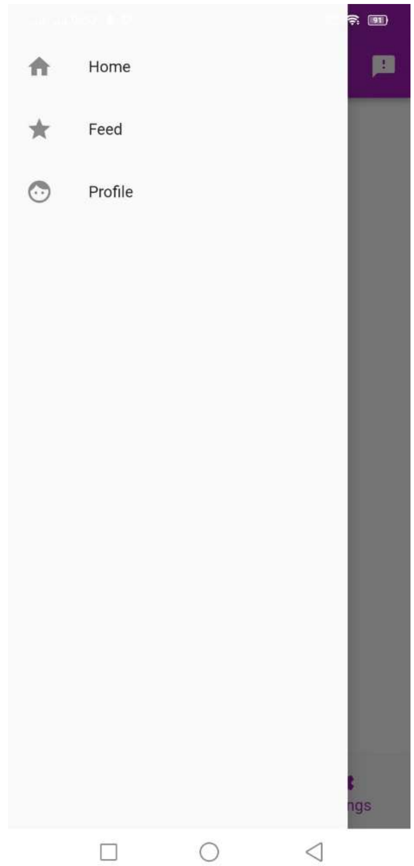

VirtualPsych
I am feeling very sad today
Tell me mor
Type a message...
Chat


VirtualPsych
do IT!
Find your inner peace and relax with guided meditation sessions.
Meditation Types
Guided Meditation
Transcendental Meditation
Yoga Meditation
Loving Kindness Meditation
Meditate


VirtualPsych
Account Settings
Notification Settings
Theme and Appearance
Privacy Settings
Sound and Vibration Settings
About
Settings


VirtualPsych
Overall Progress
Mood Tracker
Goal Tracking
Journal Entries
Sleep Tracking
Activity and Exercise Tracking
Reminders and Notifications
Insights and Analytics
Progress


4:06
Provide Feedback
Feedback
We value your feedback! Please let us know your thoughts and suggestions.
Your Feedback
Submit


Home
Feed
Profile

The *summary* the virtual psych application focuses on outlining the requirements and design considerations for the development of the app. The requirements section includes both functional and non-functional requirements that need to be fulfilled by the application.

The functional requirements specify the features and functionalities that the app must have, such as user registration and login, a chatbot for psychological support, personalized recommendations, user feedback, guidance on meditation, and progress tracking. These requirements ensure that the app provides a comprehensive and personalized user experience.

The non-functional requirements highlight the qualities and characteristics that the app must possess, including scalability to handle a large number of users, compatibility with multiple devices, and a user-friendly interface with intuitive design and navigation. These requirements contribute to the overall usability, performance, and accessibility of the app.

The chapter also provides hardware and software requirements for using the virtual psych app. These requirements specify the minimum and recommended specifications for devices, including RAM, processor, storage space, operating system, internet connection, and browser support. By meeting these requirements, users can ensure optimal performance and functionality of the app on their devices.

Furthermore, the chapter introduces the proposed methodology for the development of the VirtualPsych chatbot application. The methodology includes stages such as dataset collection, design and development of the chatbot architecture, and testing and evaluation of the app's usability and effectiveness. This methodology aims to deliver a user-friendly and accurate virtual psychologist that can provide personalized mental health support.

Lastly, the chapter provides an overview of the system architecture, which follows a client-server model. The client-side includes the frontend application and user interface, while the server-side handles the backend operations, including natural language processing, database interaction, and communication with the client. This architecture ensures smooth communication and functionality between the various components of the app.

Overall, in this chapter sets the foundation for the development and design of the virtual psych application by defining the requirements, proposing a methodology, and outlining the system architecture.

Chapter 4: Implementation and Test Cases

4.1 Implementation

During the implementation phase of our virtual psychologist chatbot app, we have made significant progress in bringing the project to life. We have focused on developing both the backend and frontend components, ensuring a robust and user-friendly application.

In the backend, we have focused on data preprocessing and model integration.

We have carefully preprocessed the conversation data by cleaning and formatting it to ensure optimal performance during training.

This involved removing noise, handling special characters, and tokenizing the text for further processing. For the core chatbot functionality, we have chosen to utilize a transformer-based model. The transformer model is a state-of-the-art architecture in natural language processing that excels in understanding and generating human-like responses.

We have integrated a transformer model into our backend using deep learning frameworks and libraries such as TensorFlow. This model will serve as the engine behind our virtual psychologist, enabling it to process user input, generate appropriate responses, and engage in meaningful conversations.

On the frontend side, we have utilized the Flutter framework to build the mobile app. We have created a login and signup page that allows users to securely register and authenticate their accounts. The login page presents a clean and intuitive interface, where users can enter their credentials and access the chatbot interface. The chatbot interface is the core component of the app, where users can engage in conversations with the virtual psychologist. We have designed an interactive and user-friendly chat interface, which dynamically displays the responses generated by the chatbot model.

Throughout the implementation phase, rigorous testing and debugging have been conducted to ensure the functionality and usability of the login/signup page and the chatbot interface. We have focused on providing a seamless user experience, adhering to Flutter's design principles and leveraging its extensive widget library.

4.1.1 Implementation of First Component/Algorithm

The first component of our virtual psychologist chatbot app focuses on implementing a machine learning algorithm to analyze user input and generate appropriate responses. While the implementation is still in progress, we have made significant advancements in developing and fine-tuning the algorithm.

For the machine learning task, we have gathered a sizable dataset of conversations between psychologists and patients, with corresponding emotional labels and response types. This dataset serves as the foundation for training and evaluating our algorithm. We are currently in the process of preprocessing the dataset, including text cleaning, tokenization, and encoding techniques to prepare the data for training.

To choose the optimal machine learning algorithm, we are exploring various options such as recurrent neural networks (RNNs), transformers, and natural language processing (NLP) models. Each algorithm has its strengths and limitations, and we are conducting thorough evaluations to identify the most suitable approach for our chatbot.

As part of the implementation, we are also incorporating sentiment analysis techniques to better understand the emotional context of user inputs. This involves leveraging pre-trained models and fine-tuning them on our specific domain to accurately detect and classify emotions expressed by users.

Additionally, we are integrating external APIs and libraries that provide advanced natural language processing capabilities. These tools assist in tasks such as entity recognition, intent detection, and sentiment analysis, enhancing the chatbot's overall understanding of user queries and enabling more accurate responses.

Throughout the development process, we are rigorously testing and evaluating the performance of the algorithm. This includes conducting validation tests, comparing different hyperparameter settings, and monitoring the model's performance on diverse test datasets. We are actively fine-tuning the algorithm to improve its accuracy, adaptability, and ability to handle various user scenarios.

As the development progresses, we are continuously gathering user feedback and conducting user acceptance testing to ensure that the chatbot meets the expectations and needs of its intended users. This feedback loop allows us to make iterative improvements and enhance the overall user experience.

In summary the implementation phase involves developing both the backend and frontend components. In the backend, data preprocessing and model integration are carried out. Conversation data is preprocessed by cleaning and formatting it, and a transformer-based model is integrated into the backend using deep learning frameworks. This model serves as the engine behind the virtual psychologist, enabling it to process user input and generate appropriate responses. On the frontend, the Flutter framework is used to build the mobile app, including a login/signup page and a chatbot interface. Rigorous testing and debugging are conducted to ensure functionality and usability. The implementation of the first component/algorithm involves gathering a dataset, preprocessing it, and exploring machine learning algorithms such as RNNs, transformers, and NLP models. Sentiment analysis techniques and external APIs are also integrated. Testing and evaluation are performed to improve the algorithm's performance, and user feedback is gathered to enhance the chatbot's user experience. The implementation process is ongoing, with a focus on developing a robust and empathetic chatbot that can effectively understand and respond to user inputs.

Chapter 5: Conclusion and Future Directions

Throughout the course of this project, we have achieved significant milestones and made valuable findings in the development of our psychologist chat bot app. Our primary objective was to create a user-friendly and empathetic platform that offers support and guidance to individuals seeking mental health assistance.

Although the implementation of natural language processing (NLP) and the chat bot functionality is still in progress, the project has achieved significant milestones. User testing has shown positive feedback regarding the app's user interface, which has been praised for its intuitive design and easy navigation. The user registration and authentication system has been successfully implemented, enabling secure account creation and personalized experiences. Another notable finding is that the app effectively organizes and presents mental health information, making it easily accessible to users.

In terms of *project scope*, we have made significant progress and covered a substantial portion. Key functionalities like the chat interface, mediation support and progress tracking have been successfully implemented. And we are working on AI model and integration with AI bot to our app.

Challenges were encountered during the project, including the collecting the dataset, and responsive app design. Despite these challenges, we successfully addressed them through extensive research and collaboration.

For future work, we recommend further enhancement and improvement of the app. Some potential areas for improvement include:

- Expanding the NLP capabilities to better understand complex user inputs and emotions.
- Strengthening data privacy measures to ensure secure storage and handling of user information.
- Integrating the app with external mental health platforms or databases to provide a more comprehensive range of resources and support options.

For FYP-2:

- Refine and optimize the existing functionalities based on user feedback and requirements.

- Implement the left-out features such as extensive progress tracking and more functions that can be helpful for our user
- Continuously update and maintain the app to ensure its compatibility with evolving technologies and user needs.
- By following this plan, future work on the project can build upon the foundation laid in FYP-1 and create a more robust and impactful psychologist chat bot app.

Overall, we are working on project to meet its objectives of creating a psychologist chat bot app that offers support and guidance to users. The app provides a user-friendly interface, empathetic responses, and valuable resource.

Chapter 6: References

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[2] N. M. D. B. D. D. D. A. a. R. G. P. Rathnayaka, "A Mental Health Chatbot with Cognitive Skills for Personalised Behavioural Activation and Remote Health Monitoring," *Sensors* , vol. 22, no. 10, p. 18, 2022.

[3] Kapoor, P., Agrawal, P., & Ahmad, Z. (2021). "Therapy Chatbot: A Relief From Mental Stress And Problems."